

Turbidity and Solar Infrared Irradiance over Maitri, Antarctica

**Pavan S. Kulkarni^{†‡}, S.L. Jain[†], B.C. Arya[†], Sachin D Ghude[†], Ashok Kumar[†],
Nazeer Ahammed[†] and Shahnawaz[‡]**

[†]Radio and Atmospheric Sciences Division, National Physical Laboratory,
New Delhi – 110012

[‡]Dept. of Physics and Astrophysics, University of Delhi,
Delhi - 110007

*Email: pavannpl@yahoo.co.in

ABSTRACT

Aerosol optical depth and solar infrared irradiance at 1020 nm using narrow band filter has been measured using a microprocessor based sun photometer during 1997, 2002, 2003 and 2004. The observations were taken on hourly basis on all clear days at Maitri, an Indian Antarctic Station (70° 46' S, 11° 45' E), during seventeenth, twenty first, twenty second and twenty third Indian Scientific Antarctic expedition. Very low AOD values and comparatively high solar infrared irradiance at 1020 nm have been observed during the entire period of observation at this clean and pristine environment of the continent. But found a clear variation in month-to-month values of AOD, which can be attributed to the local meteorological conditions.

Keywords: Aerosol optical depth, Solar infrared irradiance, Sun photometer.

INTRODUCTION

Atmospheric aerosol exerts both direct and indirect radiative forcing on climate. The attenuation of solar irradiance is mostly due to aerosols and Rayleigh scattering. Like other atmospheric constituents aerosols directly affects the radiation budget of the atmosphere by the scattering, absorption of radiation and shielding the earth against incoming solar radiation. Increasing concentrations of aerosols in the atmosphere enhance this effect and thereby produce a negative as well as positive radiative forcing. The aerosol life time varies from few days to about a week and their concentration is highly variable in space and time (Gonzalez et al., 2000). This variable aerosol concentration is strongly depend on localized source. It also has a

radiative effect of absorbing and re-emitting long wave infrared radiation. The sources of aerosols are both natural and anthropogenic. The most effective light-scattering aerosols of natural origin are sulphates formed from biogenic gaseous sulphur compounds and organic carbon formed from the partial atmospheric oxygen of gaseous biogenic organic compounds. Other natural sources include gaseous and particulate emissions from volcanoes, wildfire, and desert dust storms. The main anthropogenic source of aerosols is gaseous sulphur dioxide, oxides of nitrogen, and carbon containing gases. Major light absorbing aerosols are dominated those containing black carbon formed from the incomplete combustion of carbonaceous fuel. A subset of aerosols (size less than $0.1 \mu\text{m}$) acts as cloud condensation nuclei and thereby affects the cloud formation (Radke et al., 1989; Wigley, 1989; Charlson et al., 1992; Coakley et al., 1987). Increasing the concentration of these aerosols, indirectly, has the potential to change the microphysical, optical, and radiative properties of clouds, which, in turn, can change their reflective and absorbance properties and persistence. Aerosols play an important role in the heterogeneous chemistry and thereby in the tropical stratospheric ozone (Hofmann and Solomon, 1989) through providing surfaces for condensation and heterogeneous chemistry.

The vast ice sheet of Antarctica, is located in the southern hemisphere with no or very little human activity and has only about 2% bare land area exposed to the atmosphere (Deshpande and Kamra, 2000). The air is generally pristine and is only influenced by the long range transport of sub-micro-size particles and gaseous pollutants (Gadhavi and Jayaraman, 2004). Thus Antarctic region is a unique location for the background measurement of AOD due to the absence of any major local aerosol source and its clean environment. Aerosol studies over the sites such as Antarctica will help in estimating the background level aerosol forcing over a pristine site. Factors like high surface albedo and unique solar insolation cycle over Antarctica also makes the study an interesting one.

Aerosol optical depth describes aerosol effects on radiative transfer. It is a column integrated measurement of the relative extinction of direct sunlight due to aerosol scattering and is defined by means of the Bouguer-Lambert law. The spectral aerosol optical depth is also known as turbidity. The monitoring of aerosol optical thickness or turbidity is an important parameter for studying global climatic change. Sun photometers have widely been used to measure aerosol optical thickness and other properties of the atmosphere (Volz, 1974; Flowers, 1969; Shaw, 1983; King and Byrne,

1976). In view of the above importance, measurement of AOD and Solar radiation at 1020 nm have been carried out at Maitri, Antarctica.

EXPERIMENTAL DETAILS

Measurements of columnar ozone, water vapor and aerosol optical depth have been carried out using Microtop sunphotometer (Solar light Co., USA) at Antarctica during 1997, 2002, 2003 and 2004. Microtop sunphotometer used to measure solar radiation intensities at five wavelength bands centered around 300, 305, 312, 940 and 1020 nm with a narrow bandwidth. The total field of view of the instrument is 2.5° . Interference filters are used to select desired wavelength region and a silicon photodiode is used as the light detector. Out of the five channels, the first three filter channels at 300, 305 and 312 nm are used to derive atmospheric total column ozone and while the filter channel at 940 and 1020 are used for water vapour and aerosol optical depth.

The Microtops sunphotometer is pointed directly towards the Sun with the help of a sun-target and a pointing assembly attached to the instrument and observations of the direct solar radiation intensity in terms of voltage along with time are stored. Accuracy in time is very crucial for the estimation of the Sun position at the time of observation and hence in the computation of the optical depth. So, before starting the observation the time is set within an accuracy of few seconds, using a GPS receiver. Adverse observing conditions like clouds, haze or low sun may cause deviation of $\pm 2\%$ to $\pm 3\%$ in the measurement. The uncertainties in the aerosol optical thickness measured through sunphotometer range between 0.005 and 0.02 (Shaw, 1983; Harrison et al., 1994; Ehsani et al., 1998; Holben et al., 1998). These are due to inaccurate calibration, error in correction of molecular scattering and absorption, and instrumentation. Sun photometers are typically calibrated by using the Langley plot approach (Reagan et al., 1986; Russell et al., 1993; Schmid and Wehrli, 1995), which is based on the Beer's law. The sunphotometer was calibrated frequently as per its manufacturers schedule to take care of any system degradation.

RESULTS AND DISCUSSIONS

Antarctic continent has only two seasons and they are polar summer and polar winter. Maitri, an Indian Antarctic Station is located at the fringe of the continent at $70^\circ 46' \text{ S}$ and $11^\circ 45' \text{ E}$. At this location 24 hour day time is for around 70 days i.e., from 14th November to 28th January and 24 hour night time is for around 60 days i.e., from 27th May to 22nd July. From January 28th onwards the day time goes on reducing and night time goes on

increasing till 27th May during which the Solar Zenith angle also goes increasing day by day. Similarly 22nd July onwards the night time goes on reducing and day time goes on increasing till 14th November during which the solar zenith angle goes on decreasing day by day. The minimum solar zenith angle attended at Maitri is approximately 46° that to during the summer period. Due to this reason the instrument like sun photometer which uses Sun as the source cannot be used during the polar winter period. Therefore the entire period of observation is divided into two parts and they are pre-winter which include January to April and post-winter which include August to December. This division has been done for the better understanding and interpretation of the data and to see the seasonal effect on AOD and solar infrared radiation.

A total of four years of data set (1997, 2002, 2003 and 2004), which include around 3605 data points taken in total 526 days of observation is used in the present study. Table 1 gives a systematic view of the annual and seasonal number of days and number of observation of AOD and solar infrared radiation. Similarly in Table 2 and Table 3 gives detailed information about the annual and seasonal variation of AOD and solar infrared radiation along with $\pm$ standard deviation and $\pm$ standard error. The standard deviation and standard error given in the Table 2 and Table 3 are calculated on daily averaged values. From Table 2 and 3 it is also clear that the AOD values at Antarctica are very low and lies below 0.1 and similarly the solar infrared radiation is high with the values around 6 W/m^2 . These observations contribute to the statistics of the figures presented.

Table 1: Annual and seasonal number of days and number of observation of AOD and solar infrared irradiance

Year	Number of days			Number of observation		
	Total	Pre-winter	Post-winter	Total	Pre-winter	Post-winter
1997	150	68	82	1111	580	531
2002	120	57	63	989	436	553
2003	73	41	32	520	288	232
2004	183	78	85	985	262	723

Table 2 : Annual and seasonal mean, standard deviation and standard error for

Year	Mean(AOD)			Standard Deviation($\pm$)			Standard Error($\pm$)		
	Annual	Pre-winter	Post-winter	Annual	Pre-winter	Post-winter	Annual	Pre-winter	Post-winter
1997	0.049	0.048	0.05	0.024	0.026	0.021	0.0019	0.0032	0.0024
2002	0.063	0.062	0.064	0.03	0.033	0.026	0.0027	0.0044	0.0033
2003	0.061	0.082	0.036	0.034	0.032	0.023	0.0043	0.005	0.0042
2004	0.045	0.034	0.053	0.033	0.029	0.033	0.0024	0.0033	0.0032

Table 3 : Annual and seasonal mean, standard deviation and standard error for solar infrared irradiance

Year	Mean			Standard Deviation($\pm$)			Standard Error($\pm$)		
	Annual	Pre-winter	Post-winter	Annual	Pre-winter	Post-winter	Annual	Pre-winter	Post-winter
1997	5.9	5.88	5.92	1.01	0.85	1.13	0.083	0.104	0.125
2002	5.78	5.63	5.92	0.72	0.79	0.62	0.066	0.105	0.079
2003	6.32	6	6.75	0.59	0.49	0.43	0.07	0.076	0.078
2004	5.98	6.02	5.95	0.82	0.93	0.73	0.06	0.105	0.071

Fig.1: (a) and (b) illustrates variation of monthly averaged values of aerosol optical depth and solar infrared irradiance at 1020 nm, for the aforesaid period, over Maitri, Antarctica respectively. The vertical bars shown in the Fig.1 (a) and (b) shows the standard error in the observation set. Fig.1 (a) reveals a clear variation in month-to-month values of AOD.

The daily average values of AOD (1020 nm) for the measurement period is shown in Fig.1 (c). This observation indicates that the aerosol optical depth rarely exceeds 0.1. The AOD values are high during January and are decreasing up to April and attending low AOD. Again, August onwards levels are continuously increasing till December. The above phenomenon is cyclic in nature. Observation period adjacent to polar winter shows values

on the lower side and daily averaged AOD (1020 nm) drops below 0.02 on number of occasions in this period.

On the other hand Fig.1 (d) shows daily averaged values of solar infrared radiation at 1020 nm for the measurement period. This observation indicates that the intensity of solar infrared irradiance at 1020 nm is high and lies, almost all the time, in the range of 5.5 W/m^2 to 7.0 W/m^2 which can be seen on the burned faces of the people exposed for longer hours during sunny days at Maitri. On few occasions this values fall below 5 W/m^2 . Fig.1 (d) shows a decreasing trend in the values of solar infrared irradiance during the pre-winter period and an increasing trend in the post-winter, which can be attributed to the changing solar zenith angle and the duration of total daylight period.

Frequency histograms of AOD and solar infrared irradiance are shown in Fig.2. Fig. 2 (a) and (d) show the frequency histograms for AOD and solar infrared irradiance respectively for the complete year. Fig.2 (b) and (e) shows pre-winter and Fig.2 (c) and (f) shows post-winter, frequency histogram for AOD and solar infrared radiation respectively. The Y axis shows frequency of occurrence in percent (%) in all the histograms shown in Fig.2, where the sum of frequency is equal to 100% for each histogram. The frequency of occurrence is calculated on the number of observation given in the Table 1 and not on the daily averaged values. The aerosol optical depth at 1020 nm probability distribution is rather narrow with a modal value of about 0.05 both annually and seasonally, which indicates the pristine environment over Antarctica. In the year 1997, the post-winter probability with a modal value 0.1 is more than pre-winter time. In the year 2002, the pre and post winter frequency occurrence are equal with a modal values of 0.1. In both 2003 and 2004 years a large difference has been noticed in frequency occurrence during pre-winter and post-winter seasons with a modal value of 0.1.

In general both annually and seasonally solar infrared irradiance frequency occurrence at a modal value of 7 W/m^2 is high in compare to the other modal values.

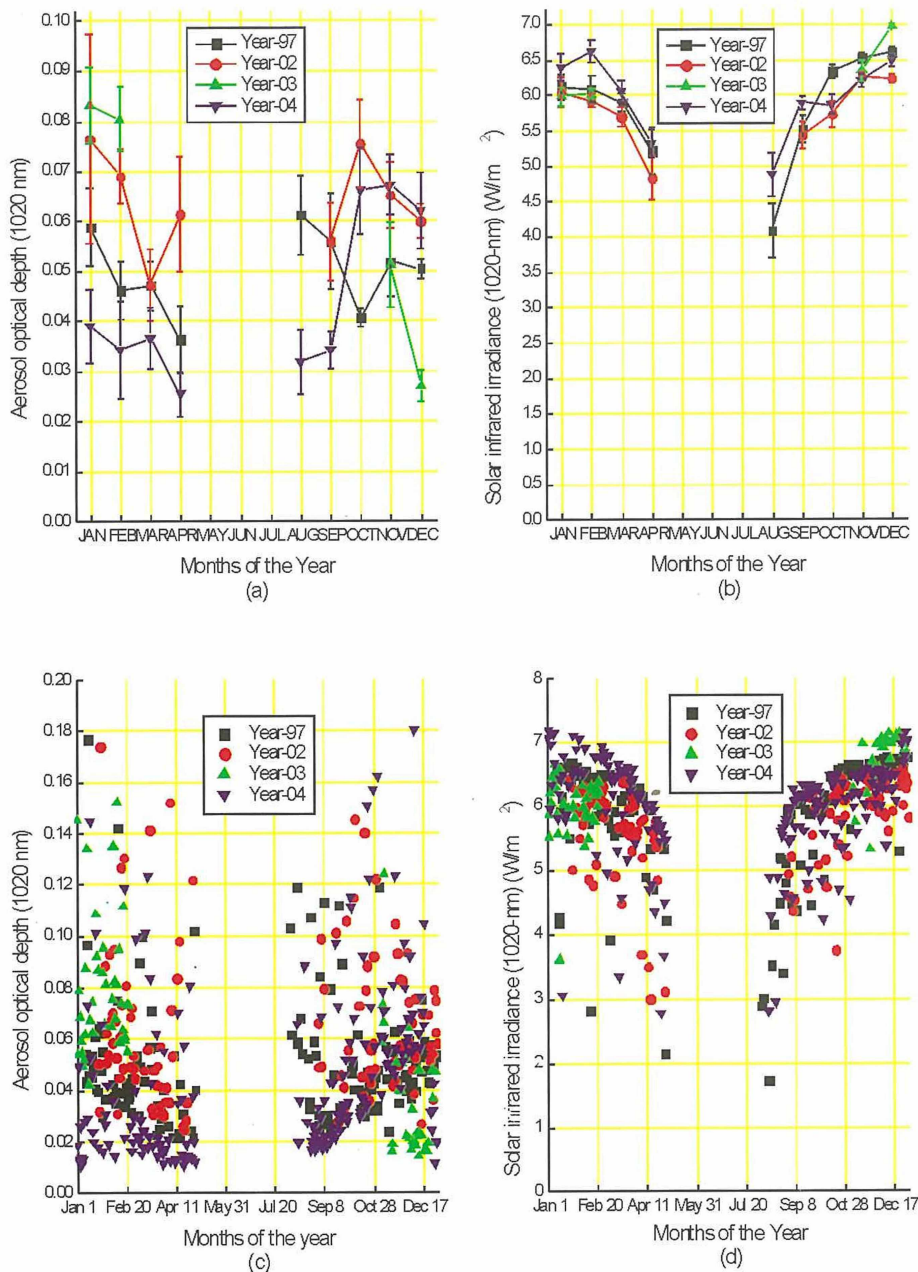


Fig.1: Monthly mean values of (a) AOD and (b) solar infrared irradiance (the vertical bars indicate $\pm$ standard error), and Daily mean values of (c) AOD and (d) solar infrared irradiance.

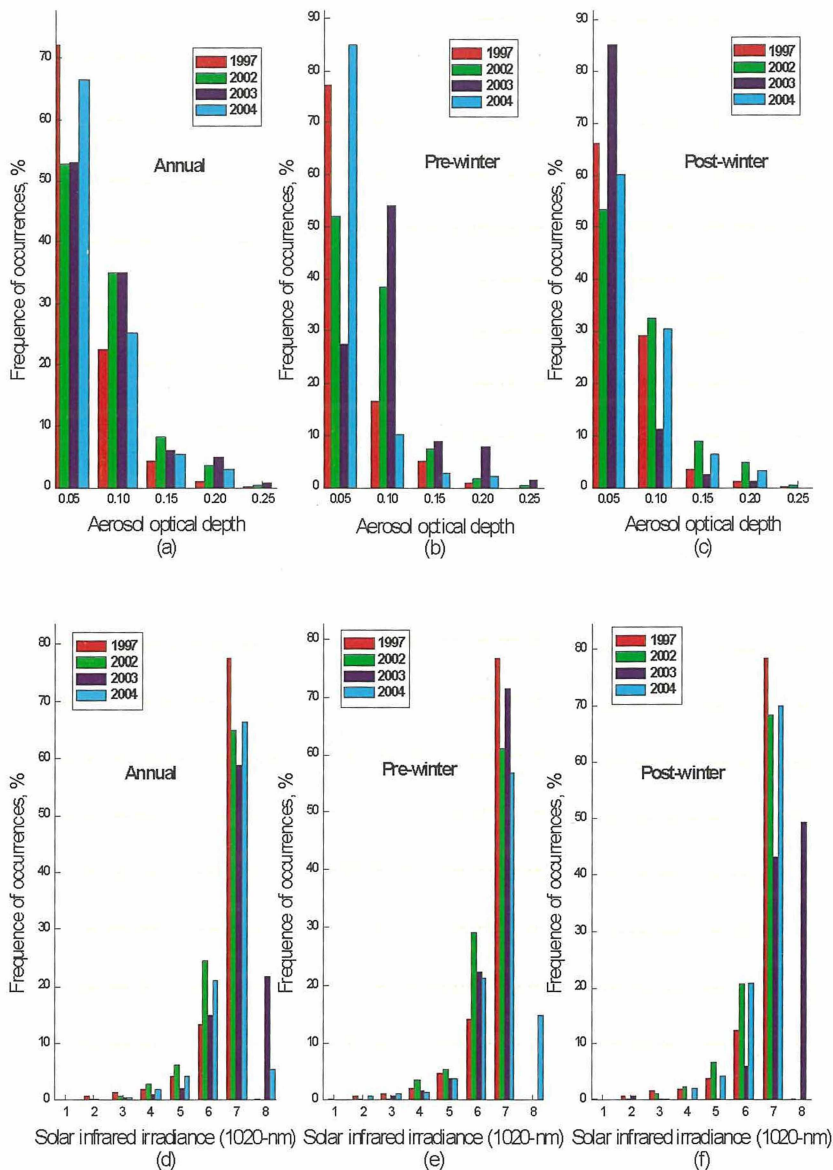


Fig. 2: Annual and seasonal frequency of occurrence of aerosol optical depth and solar infrared irradiance. The sum of the frequency is equal to the 100% for each histogram.

CONCLUSIONS

The principal conclusions drawn from present work can be summarized as follows:

- 1) This study shows that the aerosol optical depth at 1020 nm over Antarctica was very low during the period of observation. Annual and seasonal variation are negligible and can be attributed to local meteorological conditions.
- 2) Most of the time AOD at 1020 nm is less than 0.1, which shows the pristine environment over Antarctica.
- 3) Annual and seasonal solar infrared irradiance forcing is high as compared to high latitudes due to the clear sky conditions and low aerosols over this region.

ACKNOWLEDGEMENT

The authors wish to express their sincere gratitude to the Director, NPL and Head, RASD, for their encouragement during the progress of the work. Thanks are due to the NCAOR, MoES and CSIR for logistic and financial assistance. Thanks are also due to leaders of 17th, 21st, 22nd and 23rd Indian Scientific Expedition to Antarctica for providing the required help during the experiment.

REFERENCES

1. Charlson, R.J., Schwartz, S.E., Aales, J.M., Cess, R.D., Coakley Jr., J.A., Hanson, J.E., and Hofmann, B.J. (1992). Climate forcing by anthropogenic aerosol. *Science*, 255, 423-430.
2. Coakley, Jr, J.A., Bernstein, R.A., and Durkec, P.A. (1987). Effect of ship track effluents on cloud reflectivity. *Science*, 237, 1020-1022.
3. Deshpande C.G and Kamra A.K. (2000). The sub-micron aerosol size distribution measurement during the XVI Indian Antarctic Expedition. *Scientific Report, DOD*, Technical publication No. 14 (2000), 153-171.
4. Ehsani, A. R., Reagan, J.A., and Erxleben, W.H. (1998). Design and performance of an automated 10 channel solar radiometer instrument. *J. Atmos. Oceanic Technol.*, 15, 697-707.
5. Flowers, E.C., McCormick, R.A., and Kurfis, K.R. (1969). Atmospheric turbidity over the United States. *J. Appl. Meteor.*, 8, 955-962.

6. Gadhavi, H., and Jayaraman, A. (2004). Aerosol characteristics and aerosol radiative forcing over Maitri, Antarctica. *Current Science*, 86(2), 296-304.
7. Gonzalez, C.R., Deleeuw, G., Bultjes, P.J.H., Van Loon, V., and Schaap, M. (2000). Comparison between satellite retrieved aerosol optical thickness and results obtained from the lotos model. *J. Aerosol Sci.*, 31(1), 584-585.
8. Harrison, L., Michalsky, J., and Berndt, J. (1994). Automated multi-filter rotation shadow band radiometer: An instrument for optical depth and radiation measurements. *Appl. Opt.*, 33, 5118-5125.
9. Holben, B.N., and Coauthors. (1998). A federated instrument network and data archive for aerosol characterization. *Remote Sens. Environ.*, 66, 1-16.
10. Hofmann, D. J., and Solomon, S. (1989). Ozone destruction through heterogeneous chemistry following the eruption of El Chichon. *J. Geophys. Res.* 94 (D), 5029-41.
11. King, M.D., and Byrne, D.M. (1976). A method for informing total ozone content from the spectral variation of total optical depth obtained with a solar radiometer. *J. Atmos. Sci.*, 33, 2242-2251.
12. Radke, L.F., Coakley, J.A., and King, M.D.(1989). Direct and remote sensing observations of the effect of ships on clouds. *Science*, 246, 1146-1149.
13. Reagan, J.A., Thomason, L.W., Herman, B.M., and Palmer, J.M. (1986). Assessment of atmospheric limitations on the determination of the solar spectral constant from ground based spectroradiometer measurements. *IEEE Trans. Geosci. Remote Sens.*, GE 24, 258-266.
14. Russell, P.B., and coauthors. (1993). Pinatubo and pre Pinatubo optical depth spectra: Mauna Loa measurements, comparisons, inferred particle size distribution, radiation effect, and their relationship to lidar data. *J. Geophys. Res.*, 1998, 22 969- 22 985.
15. Schmid, B., and Wehrli, C. (1995). Comparison of sun photometer calibration by use of the Langley technique and the standard lamp. *Applied Optics* **34**, 4500-4512. (PMOD/WRC intern: 680)
16. Shaw, G.E. (1983). Sun photometry. *Bull. Amer. Meteor. Soc.*, 64, 4-11.
17. Volz, F.E. (1974). Economical multispectral sun photometer for measurements of aerosol extinction from 0.44 mm to 1.6 mm and precipitable water. *Appl. Opt.*, 13, 1732-1733.
18. Wigley, T.M.L. (1989). Possible climate change due to SO₂-derived cloud condensation nuclei. *Nature*, 339, 355-357.